
LAB SESSION 06

PROGRAM FLOW CONTROL USING LOOP AND NESTED LOOP

1. Write a program to display “Computer Science” ten times on computer screen.

Source Code:

```
01 TITLE LAB 6 ACTIVITY 1          ;TITLE OF THE PROGRAM
02 .MODEL SMALL                   ;DEFINE MEMORY MODEL
03 .STACK 100H                    ;SET THE SIZE OF THE STACK
04 .CODE                          ;CODE SEGMENT STARTS
05 .DATA                          ;DATA SEGMENT STARTS
06 MSG DB "M SHAHZAIB NADEEM", '$' ;DEFINE A NULL-TERMINATED STRING MSG
07 MAIN PROC                      ;START OF THE MAIN PROCEDURE
08     MOV AX,@DATA                ;LOAD THE ADDRESS OF THE DATA SEGMENT INTO AX
09     MOV DS,AX                   ;SET DATA SEGMENT DS TO THE VALUE IN AX
10
11     MOV CX,0AH                  ;INITIALIZE CX REGISTER WITH THE COUNT OF ITERATIONS
12
13 LOOP1:                          ;START OF LOOP1 LABEL
14     MOV AH,9                    ;SET THE INTERRUPT FUNCTION 'TO DISPLAY A STRING'
15     LEA DX, MSG                 ;LOAD (EFFECTIVE ADDRESS OR OFFSET ADDRESS) OF MSG INTO DX
16     INT 21H                     ;CALL INTERRUPT 21H TO DISPLAY THE STRING
17
18     MOV AH,2                     ;SET THE INTERRUPT FUNCTION 'TO DISPLAY A CHARACTER'
19     MOV DL,0DH                  ;LOAD DL REGISTER
20     INT 21H                     ;CALL INTERRUPT 21H TO DISPLAY THE CHARACTER
21
22     MOV AH,2                     ;SET THE INTERRUPT FUNCTION 'TO DISPLAY A CHARACTER'
23     MOV DL,0AH                  ;LOAD DL REGISTER
24     INT 21H                     ;CALL INTERRUPT 21H TO DISPLAY THE CHARACTER
25
26     INC BX                      ;INCREMENT THE VALUE IN THE BX REGISTER
27     CMP BX, CX                  ;COMPARE THE VALUE IN THE BX WITH THE COUNT IN CX
28
29     JL LOOP1                    ;JUMP BACK TO LOOP1 IF BX IS LESS THAN CX
30
31     MOV AH,4CH                  ;SET THE INTERRUPT FUNCTION TO EXIT THE PROGRAM
32     INT 21H                     ;CALL INTERRUPT FUNCTION 21H TO EXIT THE PROGRAM
33
34 MAIN ENDP                      ;END OF THE MAIN PROCEDURE
35 END MAIN                       ;END OF THE PROGRAM WITH THE LABEL MAIN
```

OUTPUT:

 emulator screen (80x25 chars)

```
M SHAHZAIB NADEEM
M SHAHZAIB NADEEM
M SHAHZAIB NADEEM
M SHAHZAIB NADEEM
M SHAHZAIB NADEEM
M SHAHZAIB NADEEM
M SHAHZAIB NADEEM
M SHAHZAIB NADEEM
M SHAHZAIB NADEEM
M SHAHZAIB NADEEM
M SHAHZAIB NADEEM
```

2. Write a program to display the numbers in the specific pattern as shown blow.

```

1 2 3 4 5
 2 3 4
  3
 4 3 2
5 4 3 2 1

```

Source Code:

```

01 TITLE LAB 6 ACTIVITY 2          ;TITLE OF THE PROGRAM
02 .MODEL SMALL                   ;DEFINE MEMORY MODEL
03 .STACK 100H                    ;SET THE SIZE OF THE STACK
04 .CODE                          ;CODE SEGMENT STARTS
05
06 MAIN PROC                      ;START OF THE MAIN PROCEDURE
07     MOV CX,5                    ;Initialize CX register with the count 5
08     MOV DL, '1'                 ;Initialize DL register with the character '1'
09
10 LOOP1:                          ;Start of LOOP1 label
11     MOV AH,2                    ;Set the interrupt function to display a character
12     INT 21H                     ;Call interrupt 21H to display the character
13     INC DL                      ;Increment the value in the DL register
14     LOOP LOOP1                 ;Loop back to LOOP1 until CX is not zero
15
16     MOV AH,2                    ;Set the interrupt function to 'display a character'
17     MOV DL,0DH                  ;Load DL register
18     INT 21H                     ;Call interrupt 21H to display the character
19     MOV DL,0AH                  ;Load DL register
20     INT 21H                     ;Call interrupt 21H to display the character
21     MOV DL,20H                  ;Load DL register
22     INT 21H                     ;Call interrupt 21H to display the character
23
24     MOV CX,3                    ;Initialize CX register with the count 3
25     MOV DL, '2'                 ;Initialize DL register with the character '2'
26
27 LOOP2:                          ;Start of LOOP2 label
28     MOV AH,2                    ;Set the interrupt function to 'display a character'
29     INT 21H                     ;Call interrupt 21H to display the character
30     INC DL                      ;Increment the value in the DL register
31     LOOP LOOP2                 ;Loop back to LOOP2 until CX is not zero
32
33     MOV AH,2                    ;Set the interrupt function to 'display a character'
34     MOV DL,0DH                  ;Load DL register
35     INT 21H                     ;Call interrupt 21H to display the character
36     MOV DL,0AH                  ;Load DL register
37     INT 21H                     ;Call interrupt 21H to display the character
38     MOV DL,20H                  ;Load DL register
39     INT 21H                     ;Call interrupt 21H to display the character
40
41     MOV CX,1                    ;Initialize CX register with the count 1
42     MOV DL, '3'                 ;Initialize DL register with the character '3'
43
44 LOOP3:                          ;Start of LOOP3 label
45     MOV AH,2                    ;Set the interrupt function to 'display a character'
46     INT 21H                     ;Call interrupt 21H to display the character
47     LOOP LOOP3                 ;Loop back to LOOP3 until CX is not zero
48
49     MOV AH,2                    ;Set the interrupt function to 'display a character'
50     MOV DL,0DH                  ;Load DL register
51     INT 21H                     ;Call interrupt 21H to display the character
52     MOV DL,0AH                  ;Load DL register
53     INT 21H                     ;Call interrupt 21H to display the character
54     MOV DL,20H                  ;Load DL register
55     INT 21H                     ;Call interrupt 21H to display the character
56

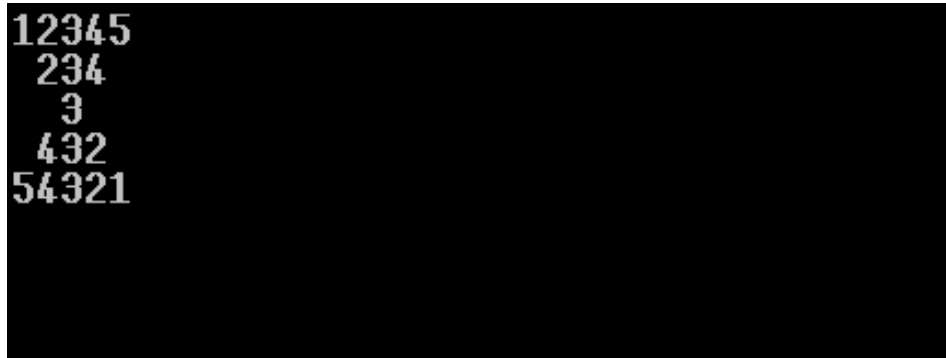
```

```

57     MOV CX,3                ;Initialize CX register with the count 3
58     MOV DL,'4'              ;Initialize DL register with the character '4'
59
60 LOOP4:                      ;Start of LOOP4 label
61     MOV AH,2                ;Set the interrupt function to display a character
62     INT 21H                 ;Call interrupt 21H to display the character
63     DEC DL                  ;Decrement the value in the DL register
64     LOOP LOOP4              ;Loop back to LOOP4 until CX is not zero
65
66     MOV AH,2                ;Set the interrupt function to 'display a character'
67     MOV DL,0DH              ;Load DL register
68     INT 21H                 ;Call interrupt 21H to display the character
69     MOV DL,0AH              ;Load DL register
70     INT 21H                 ;Call interrupt 21H to display the character
71
72     MOV CX,5                ;Initialize CX register with the count 5
73     MOV DL,'5'              ;Initialize DL register with the character '5'
74
75 LOOP5:                      ;Start of LOOP5 label
76     MOV AH,2                ;Set the interrupt function to 'display a character'
77     INT 21H                 ;Call interrupt 21H to display the character
78     DEC DL                  ;Decrement the value in the DL register
79     LOOP LOOP5              ;Loop back to LOOP5 until CX is not zero
80
81     MOV AH,4CH              ;Set the interrupt function to 'exit the program'
82     INT 21H                 ;Call interrupt 21H to exit the program
83
84 MAIN ENDP                  ; End of the main procedure
85 END MAIN                   ; End of the program with the label MAIN

```

OUTPUT:



```

12345
234
3
432
54321

```

3. Write a program to display the numbers in the specific pattern as shown blow.

```

1 2 3 4 5 6 7
1 2 3 4 5 6
1 2 3 4 5
1 2 3
1 2
1

```

Source Code:

```

01 TITLE LAB 6 ACTIVITY 3          ;TITLE OF THE PROGRAM
02 .MODEL SMALL                   ;DEFINE MEMORY MODEL
03 .STACK 100H                    ;SET THE SIZE OF THE STACK
04 .CODE                          ;CODE SEGMENT STARTS
05
06 MAIN PROC                      ;START OF THE MAIN PROCEDURE
07     MOV CX,8                   ;INITIALIZE CX REGISTER WITH THE COUNT 8
08
09 OUTERLOOP:                     ;START OF OUTERLOOP LABEL
10     PUSH CX                    ;PUSH THE CURRENT VALUE OF CX ONTO THE STACK
11     MOV DL,31H                 ;INITIALIZE DL REGISTER WITH THE CHARACTER '1' (ASCII CODE 31H)
12
13 INNERLOOP:                     ;START OF INNERLOOP LABEL
14     MOV AH,2                   ;SET THE INTERRUPT FUNCTION TO 'DISPLAY A CHARACTER'
15     INT 21H                   ;CALL INTERRUPT 21H TO DISPLAY THE CHARACTER
16     INC DL                     ;INCREAMENT THE VALUE IN THE DL REGISTER
17     LOOP INNERLOOP            ;LOOP BACK TO INNERLOOP UNTIL CX IS NOT ZERO
18
19     MOV AH,2                   ;SET THE INTERRUPT FUNCTION TO 'DISPLAY A CHARACTER'
20     MOV DL,0AH                 ;LOAD DL REGISTER
21     INT 21H                   ;CALL INTERRUPT 21H TO DISPLAY THE CHARACTER
22     MOV DL,0DH                 ;LOAD DL REGISTER
23     INT 21H                   ;CALL INTERRUPT 21H TO DISPLAY THE CHARACTER
24
25     POP CX                     ;POP THE VALUE FROM THE STACK BACK INTO CX
26     LOOP OUTERLOOP            ;LOOP CACK TO OUTERLOOP UNTIL CX IS NOT ZERO
27
28     MOV AH,4CH                 ;SET THE INTERRUPT FUNCTION TO EXIT THE PROGRAM
29     INT 21H                   ;CALL INTERRUPT 21H TO EXIT THE PRPGRAM
30
31 MAIN ENDP                      ;End of the main procedure
32 END MAIN                      ;End of the program with the label MAIN

```

OUTPUT:

 emulator screen (80x25 chars)

```

12345678
1234567
123456
12345
1234
123
12
1

```

4. Write a program to display number “1” on the screen and then after each 60 second update it.

Source Code:

```

01 TITLE LAB 6 ACTIVITY 4                ;Title of the program
02 .MODEL SMALL                          ;Dedine memory model
03 .STACK 100H                           ;Set the size of the stack
04 .CODE                                 ;Code segment starts
05
06 MAIN PROC                             ;Start of the main procedure
07     MOV CX,9                           ;Initialize CX register with the count 9
08     MOV DL,'1'                         ;Initialize DL register with the character '1'
09
10 LOOP1:                                ;Start of LOOP1 label
11     MOV AH,2                           ;Set the interrupt function to display a character
12     INT 21H                             ;Call interrupt 21H to display the character
13
14     PUSH DX                             ;Push the value in DL onto the stack
15     MOV AH,2                           ;Set the interrupt function to display a character
16     MOV DL,8                           ;Load DL register with the backspace character (ASCII code 8)
17     INT 21H                             ;Call interrupt 21H to display the backspace character
18
19     POP DX                              ;Pop the value from the stack back into DL
20     INC DL                              ;Increment the value in the DL register
21
22     PUSH CX                             ;Push the value in CX onto the stack
23     MOV CX,0BFFH                       ;Initialize CX register with a (60 second) count
24
25 LOOP2:                                ;Start of LOOP2 label
26     NOP                                ;No operation (placeholder instruction)
27     LOOP LOOP2                          ;Loop back to LOOP2 until CX is not zero
28
29     POP CX                              ;Pop the value from the stack back into CX
30     LOOP LOOP1                          ;Loop back to LOOP1 until CX is not zero
31
32     MOV AH,2                           ;Set the interrupt function to display a character
33
34 MAIN ENDP                              ;End of the main procedure
35 END MAIN                               ;End of the program with the label MAIN

```

OUTPUT:

 emulator screen (80x25 chars)

```

1
2
3
4
5
6
7
8
9

```